

Cambridge (CIE) IGCSE Chemistry
Paper 4: Theory (Set D)
Extended

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

/ 82

Instructions

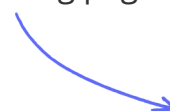
- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- You should show all your working and use appropriate units.

Materials

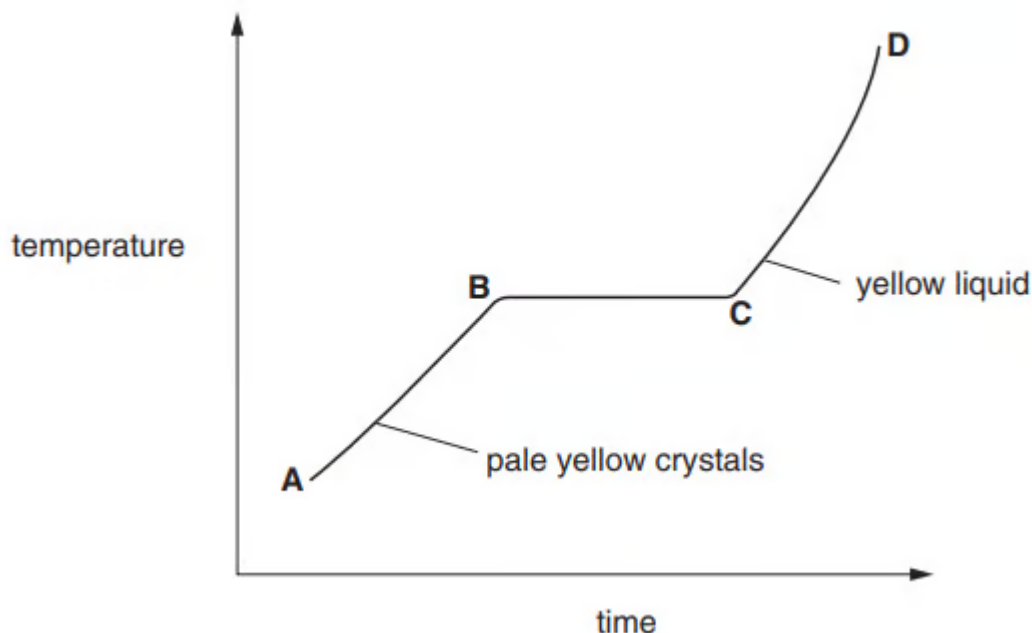
- A copy of [the Periodic Table of elements](#) is required for this paper.

Scan here to mark your mock exam

or visit the mock exams landing page for this course



- 1 (a) When nitrogen dioxide is cooled, it forms a yellow liquid and then pale yellow crystals. These crystals are heated and the temperature is measured every minute. The following graph can be drawn.



- i) Describe the arrangement and movement of the molecules in the region A-B.
- ii) Name the change that occurs in the region B-C

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(4 marks)

(b) Nitrogen dioxide and other oxides of nitrogen are formed in car engines.

i) Explain how these oxides are formed.

ii) How are they removed from the exhaust gases?

(4 marks)

(c) Nitrogen dioxide, oxygen and water react to form dilute nitric acid.

Describe how lead(II) nitrate crystals could be prepared from dilute nitric acid and lead(II) oxide.

(3 marks)

2 (a) Protons, neutrons and electrons are subatomic particles.

Complete the table to show the relative mass and relative charge of a proton, a neutron and an electron.

particle	relative mass	relative charge
proton		
neutron		
electron	$\frac{1}{1840}$	

[3]

(3 marks)

(b) Bromine has two isotopes.

i) Define the term isotope.

[2]

ii) Explain why the two isotopes of bromine have the same chemical properties.

[2]

(4 marks)

(c) The table shows the number of protons, neutrons and electrons in some atoms and ions.

Complete the table.

particle	number of protons	number of neutrons	number of electrons
${}^7_3\text{Li}$			
${}^{34}_{16}\text{S}^{2-}$			
	19	22	18

(5 marks)

3 (a) Carbon dioxide and silicon(IV) oxide are oxides of Group IV elements.

Complete the following table.

	carbon dioxide	silicon (IV) oxide
formula		SiO ₂
melting point / °C	-56	1610
physical state at 25 °C	gas	
conduction of electricity	non-conductor	
structure		macromolecular

(4 marks)

(b) i) Name the type of bonds that exist between the atoms in silicon(IV) oxide.

[1]

ii) Explain why silicon(IV) oxide has a very high melting point.

[1]

iii) Explain, in terms of attractive forces between particles, why carbon dioxide has a very low melting point.

[1]

iv) Explain, in terms of particles, why carbon dioxide is a non-conductor of electricity.

[1]

(4 marks)

(c) i) Name the type of chemical reaction in which carbon dioxide is produced from fossil fuels.

[1]

ii) Name the chemical process in which green plants convert carbon dioxide into carbohydrates.

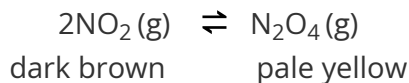
[1]

(2 marks)

4 (a) Nitrogen dioxide is a brown gas. It can be made by heating certain metal nitrates.



At most temperatures, samples of nitrogen dioxide are equilibrium mixtures.



i) At 25 °C, the mixture contains 20 % of nitrogen dioxide. At 100 °C this has risen to 90%. Is the forward reaction exothermic or endothermic? Give a reason for your choice.

[2]

ii) Explain why the colour of the equilibrium mixture becomes lighter when the pressure on the mixture is increased.

[2]

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(4 marks)

- (b) A 5.00 g sample of impure lead(II) nitrate was heated. The volume of oxygen formed was 0.16 dm³ measured at r.t.p. The impurities did not decompose.

Calculate the percentage of lead(II) nitrate in the sample.



Number of moles of O₂ formed =

Number of moles of Pb(NO₃)₂ in the sample =

Mass of one mole of Pb(NO₃)₂ = 331 g

Mass of lead(II) nitrate in the sample = g

Percentage of lead(II) nitrate in sample =

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(4 marks)

5 (a) Ammonia is a compound which only contains the elements nitrogen and hydrogen. It is a weak base.

i) Define the term *base*.

[1]

ii) Given aqueous solutions of ammonia and sodium hydroxide, both having a concentration of 0.1 mol / dm^3 , how could you show that ammonia is the weaker base?

[2]

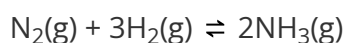
(3 marks)

(b) Ammonia is manufactured by the Haber Process. The economics of this process require that as much ammonia as possible is made as quickly as possible.

Explain how this can be done using the following information.

The conditions for the following reversible reaction are:

- 450°C
- 200 atmospheres pressure
- iron catalyst

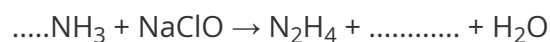


the reaction is exothermic

(5 marks)

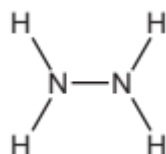
- (c) Another compound which contains only nitrogen and hydrogen is hydrazine, N_2H_4 .

Complete the equation for the preparation of hydrazine from ammonia.



(2 marks)

- (d) The structural formula of hydrazine is given below.



Draw a diagram showing the arrangement of the valency electrons in one molecule of the covalent compound hydrazine.

Use x to represent an electron from a nitrogen atom.

Use o to represent an electron from a hydrogen atom.

(3 marks)

(e) Hydrazine is a weak base and it removes dissolved oxygen from water. It is added to water in steel boilers to prevent rusting.

i) One way it reduces the rate of rusting is by changing the pH of water. What effect would hydrazine have on the pH of water?

[1]

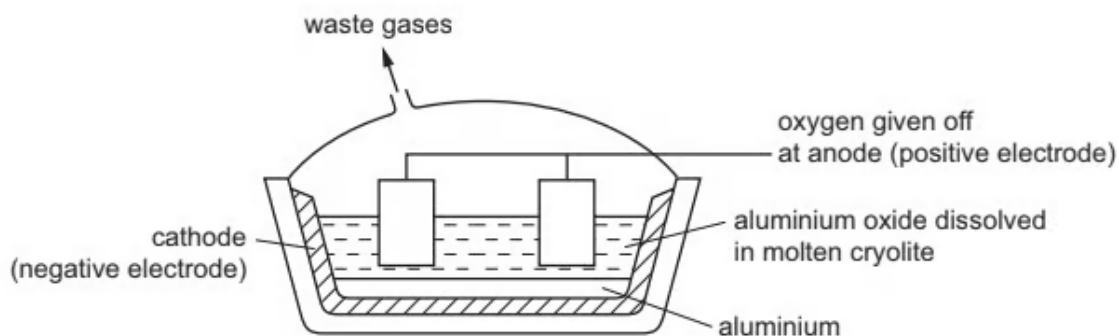
ii) Give a reason, other than pH, why hydrazine reduces the rate of rusting.

[1]

(2 marks)

6 (a) Aluminium and iron are extracted from their ores by different methods.

Aluminium is extracted from its purified oxide ore by electrolysis.



What is the name of the ore of aluminium which consists mainly of aluminium oxide?

(1 mark)

(b) The electrodes are both made of the same substance.

Name this substance.

(1 mark)

(c) Aluminium oxide is dissolved in molten cryolite before it is electrolysed.

Give two reasons why aluminium oxide dissolved in molten cryolite is electrolysed rather than molten aluminium oxide alone.

(2 marks)

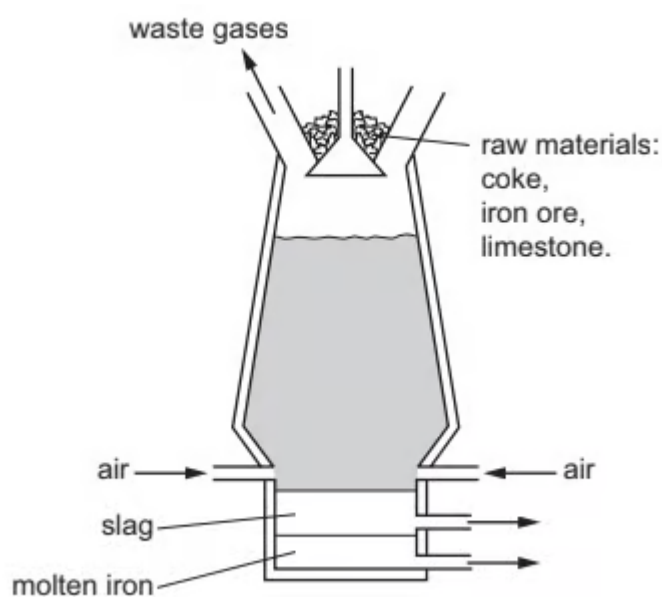
(d) Write the ionic equations for the reactions at the electrodes in this electrolysis.

Anode (positive electrode)

Cathode (negative electrode)

(2 marks)

(e) Iron is extracted from its oxide ore by reduction using carbon in a blast furnace.



i) Place the elements aluminium, carbon and iron in order of reactivity with the least reactive element first.

[1]

ii) Use your answer to (e)(i) to explain why iron is extracted by reduction using carbon but aluminium is not.

[1]

(2 marks)

(f) What is the name of the ore of iron which consists mainly of iron(III) oxide?

(1 mark)

7 (a) Use your copy of the Periodic Table to answer these questions.

Choose an element from the Periodic Table to match each description. You may give either the name or the symbol.

i) It is the most reactive metal.

[1]

ii) It is the only non-metal which is a liquid at r.t.p.

[1]

iii) This Group VII element is a solid at r.t.p.

[1]

iv) This element is in Group V and Period 4.

[1]

v) This unreactive gas is used to fill lamps.

[1]

(5 marks)

(b) Predict the formula of each of the following compounds.

i) Germanium oxide

[1]

ii) Tellurium bromide

[1]

(2 marks)

(c) Give the formula of each of the following ions.

i) Strontium

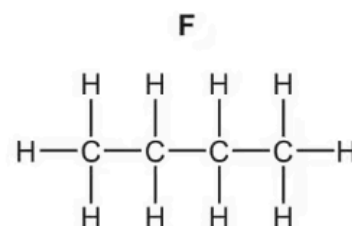
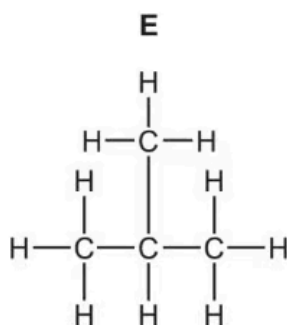
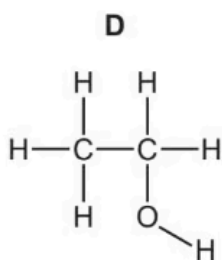
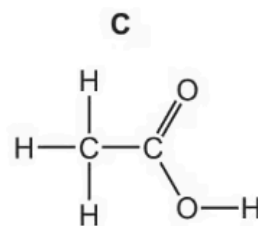
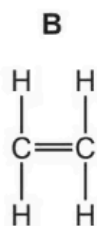
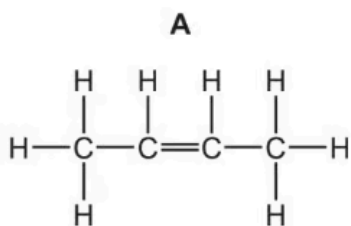
[1]

ii) Fluoride

[1]

(2 marks)

8 (a) The structures of six organic compounds are shown.



Give the name of **F**.

(1 mark)

(b) Identify **two** of the compounds that are members of the same homologous series.

Give the general formula of this homologous series.

compounds

general formula

(2 marks)

(c) Which **two** compounds are isomers of each other?

Explain why they are isomers.

compounds

explanation

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(3 marks)

(d) Explain why **B** is an unsaturated hydrocarbon.

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(2 marks)